Universität Zürich[UZH]

GEO 877 – Spatial Algorithms

Chatty Maps: Constructing a Sound Map of Zurich from Flickr Data

Aiello et al. (2016)

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1. Introduction

Previous studies have shown that noise pollution, especially in cities can have multiple effects on human health, development and stress levels (Aiello et al. 2016). Aiello et al. (2016) describe a significant problem when dealing with noise pollution data: Most cities collect the noise data at specific points all over the city. This may result in missing spatial resolution of the output and focuses mainly on negative noise pollution caused by traffic and other factors (Aiello et al. 2016). However, it does not include sounds coming from nature, music or humans and therefore does not capture the whole sound sphere.

Based on the ISO 12913, soundscapes are an "acoustic environment as perceived or experienced and/or understood by the individual, or by a society" (International Organization for Standardization, 2014). This includes not only traffic noise, but all the noises combined in an environment (Schafer et al., 1993). Because soundscapes are based on human perception, they are also dependent on how people think and categorize sounds (Davies et al. 2013). Social media data like Flickr posts can therefore reflect the perception as well as the categorization of the perceived sounds.

Based on Aiello et al. (2016) this project aims to retrieve sound information from social media data. Flickr posts and their tags are used to analyse the soundscapes of Zurich, Switzerland. As a result of this study a sound map was generated, combining visual and acoustic approaches for a richer representation of the sounds that shape Zurich (Thulin, 2018).

2. Motivation

Our project work is based on the paper from Aiello et al. (2016). The main workflow for the generation of the sound wheel and the map that shows the soundscape of the entire city was adapted to Zurich, Switzerland. Like Aiello et al. (2016) the tags of Flickr pictures were used to collect the sound information.

We chose the city area of Zurich as our main area of interest, because being a touristic destination lots of Flickr data could be retrieved from the dataset. In Zurich, as well as in other cities mostly noise pollution data at certain points gets collected while other information about sounds, especially from music, humans or the surrounding nature, gets lost. The city with its surrounding nature and hills like Uetliberg and parks like Irchelpark, but also with its rich cultural scene provides a fitting study case for the exploration of additional sounds beside noise pollution.

This approach focuses mainly on the sound related part of the Aiello et al. (2016) paper. The authors also included an emotional analysis of the Flickr tags as well as a perceptual layer, which was not used in this case. Further, the comparison with actual noise pollution data was not conducted.

3. Methods

The main categories of the sound wheel are based on the categories defined by Aiello et al. (2016). The words connected to each sound category were chosen based on looking at the Flickr posts but also by using the words provided in the paper. Zurich, like the cities of Barcelona and London studied by Aiello et al. (2016), is a large Western European and international city, with traffic noise as well as a tourist destination with green spaces and lots of cultural events. For this reason, it was assumed that most words would apply as they did for the other cities. However, some specific Swiss German words like “tram” were added to the list.

It was tried to implement a translation of the different Flickr tags but due to the number of unique posts (around 78 000 for the area of Zurich), we decided against it to avoid long processing. Instead, different words and versions of words were included in the sound dictionary, both in English and German.

The street network geodatabase comes from Swisstopo and uses the Swiss coordinate system EPSG:2056 (CH1903+ LV95), in which distances are already measured in metres. Each street is stored as a line built from a sequence of connected points, together with its name. To reduce the file size, the streets were prefiltered for Zurich in QGIS and exported as a shapefile, then cut into segments of at most 1000 metres so that each piece can later carry its own sound profile.

For each street segment longer than 100 metres, a buffer polygon with a fixed width of 25 metres is generated. The filtering of street segments and the omission of semicircular end caps in the buffer construction were implemented to reduce buffer overlaps and to ensure that only relevant segments were included in the analysis. The buffer construction was based on offsetting the segment geometry and accounting for changes in direction at vertices through the creation of curved transitions. These buffers were then used for the spatial matching of Flickr observations to street segments.

A point-in-polygon procedure based on ray casting was used to spatially match Flickr observations to buffered street segments. Prior to the analysis, all Flickr coordinates were reprojected into the street network coordinate system (EPSG:2056). For each buffer polygon, photos located within its boundaries were identified and their sound categories aggregated. The most frequent category was then assigned as the dominant sound type of the corresponding street segment.

4. Results

Our results show that 46’128 Flickr photos had at least one sound tag. Looking at the segment level, from the 2851 remaining segments after dropping the ones shorter than 100 metres, 954 segments could be matched with at least one sound. This means that 33.5% of the segments could be assigned a dominant sound category, while the remaining segments lacked sufficient sound-related observations. For all the segments analysed, the dominant categories are listed in Table 1.

The sound map (Figure 1) shows the distribution of the matched segments in Zurich. The transport segments (red) appear throughout the city and form the largest network, especially along major roads. Nature segments (green) are more commonly distributed towards the outer parts of the city. This pattern is consistent with the presence of forests (Uetliberg, Käferberg), parks and recreational areas at the edges of the city centre.

Indoor (blue) and music (purple) segments appear denser in central areas and residential areas, while mechanical (grey) and human (orange) segments only occur sporadically. The occurrence of music segments is connected to areas associated with cultural activities, such as music and arts institutions, as well as districts with a high concentration of bars, clubs and nightlife.

Table 1: Dominant Sound Category and Number of assigned Segments.

DOMINANT SOUND CATEGORY	SEGMENTS ASSIGNED TO CATEGORY
None	1897
Transport	387
Nature	307
Indoor	127
Music	57
Human	42
Mechanical	34

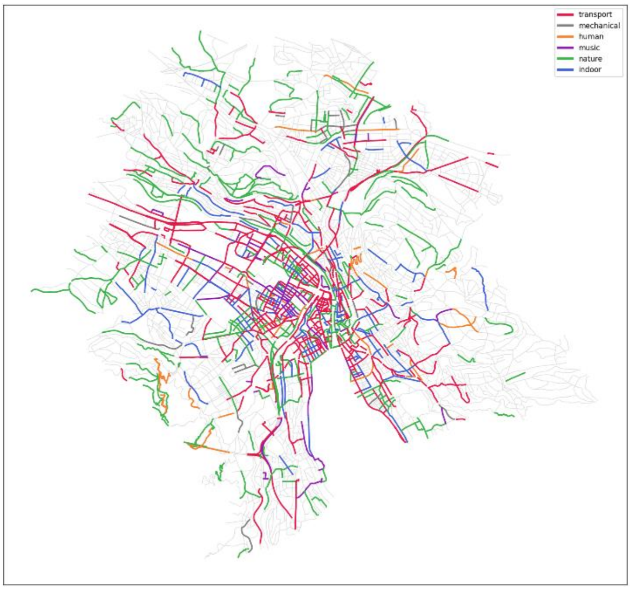

Figure 1: Sound Map for the City of Zurich.

5. Discussion and Limitations

Flickr photos are used as a proxy for soundscapes and people photograph what looks interesting rather than what they hear. A quiet but attractive square draws many photos while a busy motorway draws almost none. That is why the soundscape recovered here is really the soundscape of photogenic places. This bias sits underneath every later step. It is also the same assumption made in the original Aiello et al. (2016) study, so the approach is well established.
Further, Flickr coverage is uneven and concentrates in the old town, along the lake and in tourist areas, which is why only 954 of the roughly 2851 kept segments received a dominant sound. Residential and industrial parts of the city are barely sampled. The tags are written by users and are noisy, they mix German and English and a word such as "street" may simply be a location label rather than a sign of traffic. Geotags can also be inaccurate, so with a buffer of only 25 meters a photo can be attached to the wrong street.
The taxonomy is a further source of uncertainty. The dictionary was partly built by hand and reflects our own judgement of which words belong to which sound. Also, the dominant category is then decided by a simple majority, so a segment with three transport hits and two nature hits is labelled transport without any record of how narrow that margin was.
Several method choices were made without calibration. The segment length of 1000 meters, the buffer width of 25 meters and the twelve point approximation of the rounded corners were all fixed in advance. A segment of one kilometre can cross quite different surroundings and one fixed width treats a wide motorway and a narrow alley in the same way. Because buffers overlap near junctions, a single photo can be counted for several segments at once instead of being assigned to the closest street.
Finally, the output was never validated against real measurements. The map looks geographically plausible, but it was not compared with any acoustic recordings.

6. Credit Information

Lea did the preprocessing of the Flickr data and created the sound wheel. She also wrote Introduction, Motivation and Results of the paper. Julia developed the first algorithm, the segmentation of the streets and documented it in the report. She further wrote the discussion and limitation part of the paper. Muriel developed the buffer algorithm and documented it in the report.

7. Description of Algorithms

Algorithm 1: Segmentation of Streets

The first algorithm cuts the long street polylines into shorter segments so that each one can later be coloured by its dominant sound profile. The algorithm is built on two small geometry classes, Point and Polyline, which are taken from the lecture and shared by all three algorithms. A Point holds an x and y coordinate and knows how to measure the straight-line distance to another point. A Polyline is a line made from a sequence of connected points. When a Polyline is built, it removes any duplicate points that sit directly next to each other, so the geometry stays clean before any cutting happens.

Three functions on the Polyline implement the segmentation, each operating on distance measured along the line. The first one walks along the line and adds up the length of every edge to get the total length using Euclidean distance. The second one finds the point that sits at a given distance from the start. It moves edge by edge until it reaches the edge that contains the target distance, then places the new point on that edge by linear interpolation. The third function uses these two to cut out a sub-polyline between a start distance and an end distance. It keeps every original vertex that falls inside that range and adds interpolated points at the two ends, so the real shape of the street is preserved.

The splitting itself is straightforward. Starting at zero distance, the algorithm takes steps of 1000 metres along the line. Each step produces one segment and the walk continues until the end of the street is reached. The 1000 metre limit keeps the segments short enough to describe a local soundscape while avoiding an unnecessary flood of tiny pieces.

The main loop reads the Zurich street shapefile in EPSG:2056, where distances are already metres. It handles both single lines and multi-part lines, builds a Polyline for each one and splits it into segments. Every segment is stored together with its street name, an identifier, its length in metres and its coordinates. Running this over the full network turns the 2480 streets into roughly 4400 segments, which form the input for the buffer step that follows.

Algorithm 2: Buffer

The second spatial algorithm constructs a buffer polygon around each street segment. The buffer is built by placing offset points to the left and right of the polyline at a perpendicular distance equal to half the desired buffer width.

For each pair of consecutive points on the street segment, the direction of the segment is described by its unit and normal vector. The normal vector is used to calculate the coordinates of the offset points based on the buffer radius. Points on the right side are offset in the opposite direction of the normal vector. For the first and last point of the street segment, two parallel

offset points are computed using the normal vector of the first segment and stored in two separate lists, one for the left side and one for the right side.
For all intermediate points, a turn test is conducted to determine if there is a left, right or no turn. On the outer side of a turn, the function calculates the circular arc based on the difference of the direction of the incoming and to outgoing angle. This enables the determination of the rotational path. Along this path, the defined number of points is placed at equal angular intervals. On the inner side of a turn the coordinates of only one point are determined. Its coordinates are obtained from the intersection of the extended inner offset lines of the incoming and outgoing segments. If the turn test results no turn (t =0), two lines of the street segments are parallel. In this case a simple offset point is placed on each side, identical to construction of the start and end caps. Once all points have been collected, the polygon is assembled by reading the left-side list in its original order and appending the right-side list in reverse. This ensures that the outline traces the polygon continuously in one direction. The first point is then appended again at the end to explicitly close the polygon.

Algorithm 3: Point in Polygon Ray Casting

The third algorithm decides which photos fall inside each street buffer, so that every segment can be assigned to the sound categories of the photos around it. Because the streets are stored in EPSG:2056 while the Flickr photos arrive in WGS84 degrees, the photos are first reprojected into the street coordinate system, so that points and buffers are measured on the same metric grid.
The test at the centre of the algorithm is point-in-polygon by ray casting, following the even-odd rule from the lecture. The idea is to send a horizontal ray from the photo towards the right and count how many edges of the buffer it crosses. An even number of crossings means the photo lies outside the polygon and an odd number means it lies inside. The method proceeds in four steps. A bounding box check comes first, since a point outside the box of the polygon cannot be inside the polygon itself and can be rejected at once. The ray is then cast, the crossings are counted and the even-odd rule gives the final decision.
Two special cases need care. Horizontal edges run along the ray and are skipped, because they do not give a clean crossing. Crossings are also counted with a half-open rule on the vertical band of each edge, where the lower endpoint is included and the upper one excluded, so that a ray passing exactly through a shared vertex is counted once rather than twice.
The matching loop applies this test buffer by buffer. For each buffer, the full set of photos is first screened against the buffer's bounding box and only the points inside that box go on to the full ray casting test. Every photo found inside contributes its sound categories and the most frequent main category becomes the dominant sound of the segment. Run across the network, this leaves 954 segments with a dominant sound, which are the ones drawn in colour on the final map.

8. References

8.1 Image Sources

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8.2 Literature

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Davies, W. J., Adams, M. D., Bruce, N. S., Cain, R., Carlyle, A., Cusack, P., ... & Poxon, J. (2013). Perception of soundscapes: An interdisciplinary approach. *Applied acoustics*, *74*(2), 224-231.

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9. AI statement

In this project AI was used for following purposes:

- Checking program codes with Claude (Anthropic)
- Bug-fixing in the Code with ChatGPT (Version 5.3-mini)
- Grammar and spelling checking with Claude (Anthropic)

We confirm that we have checked the accuracy of the AI-generated content to the best of our knowledge and beliefs and we declare that we personally take full responsibility for this human-machine collaboration outcome.